

CS 57800 – Statistical Machine Learning

Lecture 0 (08/24)

https://ruizhezhang.com/course_fall_2026.html

Course Staff

Ruizhe Zhang

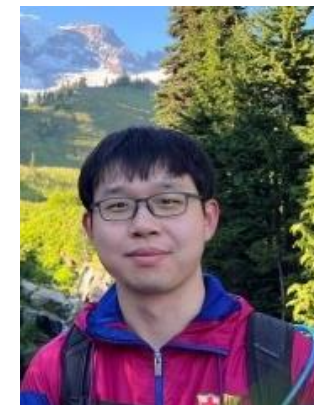
- Assistant Professor, Purdue CS
- Research: Theory—Quantum Computing, Optimization, ML Theory
- **Office hours: Monday 10:30 AM - 11:30 AM at DSAI 1119B**

Teaching assistants



Shreya Nasa

**Office hours:
Thu 9:00 AM – 10:00 AM
at DSAI B061**



Yucheng Zhang

**Office hours:
Thu 3:00 PM – 3:40 PM
at DSAI B061**

Outline

- Logistics
- Motivations
- Intro to Learning Theory

Course Instruction

- Course website: https://ruizhezhang.com/course_fall_2026.html
 - Slides and additional references
- Brightspace: <https://purdue.brightspace.com/d2l/home/1631642>
 - Announcements and problem sets submissions
- Ed Discussion for discussions
 - Access via the link in Brightspace

Grading

- **30% Problem sets**
 - 8 problem sets in total and 4 points for each
 - Must be typed in LaTeX and submit the PDFs
 - I may randomly choose problems for grading. Full solutions will be posted at Brightspace
- **30% Two midterms**
 - 15% each; in-class, closed-book, and pen-and-paper
 - Dates will be announced later
- **40% Final project**

Final Project

- **Goal:** Gain first-hand experience with (theoretical) machine learning research and develop basic scientific research skills
- **Theme:** Investigate a research question in statistical machine learning not answered by the literature, and make a focused, technically substantive, verifiable contribution
- **Collaboration:** Work in teams of two or three students
- **Timeline:**
Proposal → Mid-semester check-in (ungraded) → Final report → Presentation



AI Policy for This Course

- Generative AI is **welcome as a research tool** throughout this course
- You remain responsible for **every claim, citation, proof, computation, figure, and line of code** you submit
- Your submissions should include a brief statement of **how AI was used** and **how its outputs were checked**

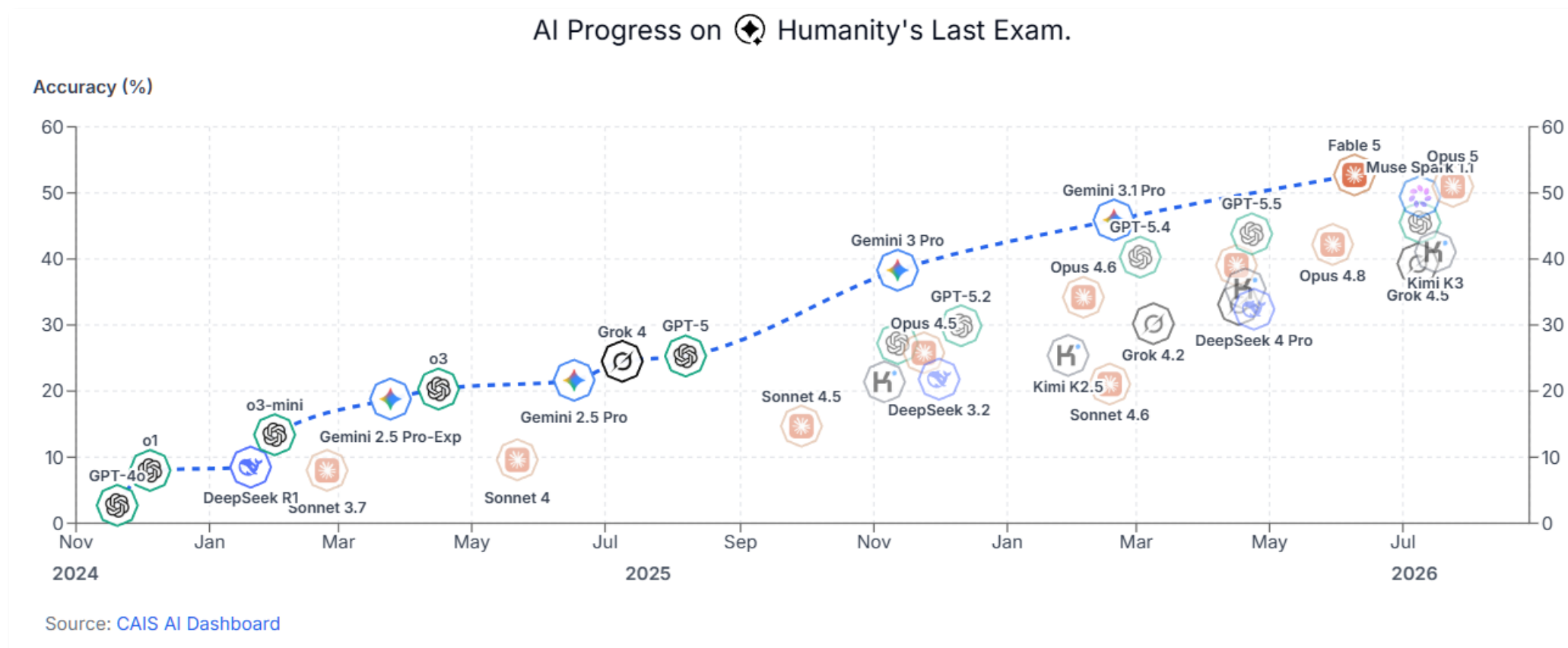


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Why Statistical Machine Learning in the LLM Era?

“Frontier models can implement, tune, and explain every algorithm—why spend a semester learning the ML theory & algorithms?”



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Start by Conceding the Premise

Frontier models can already ...

- implement standard learning algorithms
- generate experiments and visualizations
- tune models and hyperparameters
- explain concepts and proofs in textbooks

So only learning how to **implement** the algorithms is no longer enough

Start by Conceding the Premise

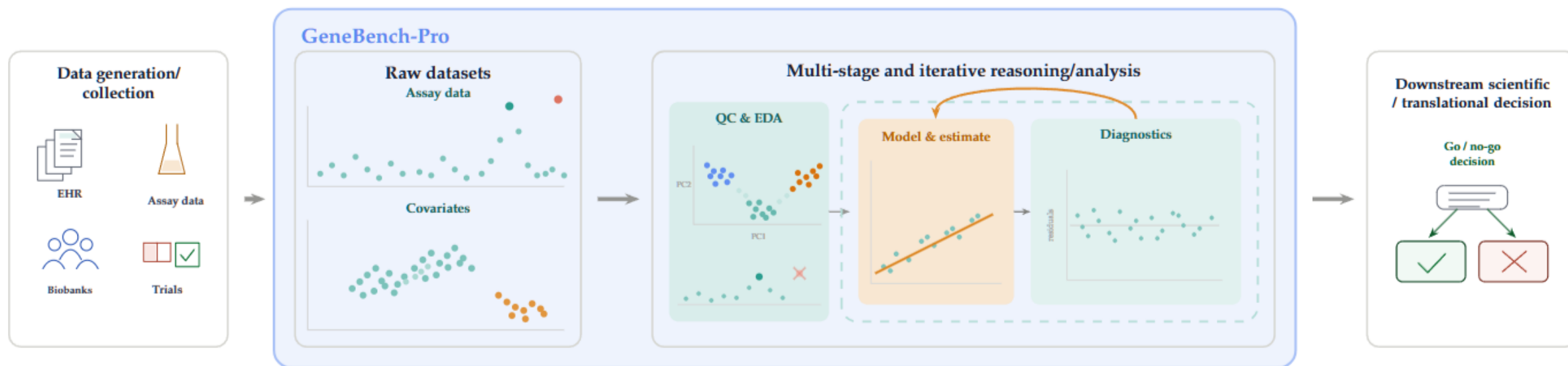
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GeneBench-Pro: Evaluating Multistage Statistical Reasoning in Genomics, Quantitative Biology, and Translational Biomedicine

Jeremy Li[†], Andrew Ho[†]
OpenAI

June 30, 2026



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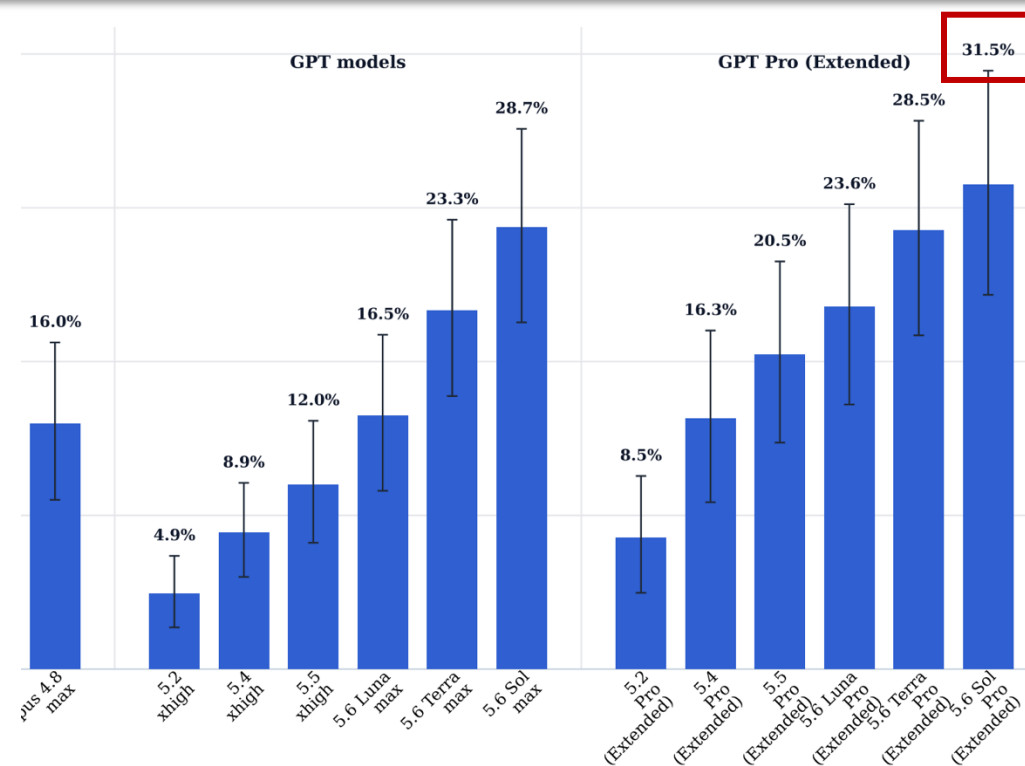
So only learning how to **implement** the algorithms is no longer enough

The scarce skill is moving from implementation to **formulation, evaluation, diagnosis, and judgment**

GeneBench-Pro: Evaluating Multistage Statistical Reasoning in Genomics, Quantitative Biology, and Translational Biomedicine

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There Is No Task-Independent “Best Classifier”

“Best” depends on the decision problem:

- deployment population and sampling process
- meaning of the label and available information
- loss function and asymmetric error costs
- privacy, latency, and computational constraints

Cost-sensitivity Bayes rule:

$$f_{\star}(x) = \mathbf{1} \left\{ \eta(x) \geq \frac{c_{\text{FP}}}{c_{\text{FP}} + c_{\text{FN}}} \right\}$$

Models optimize specified objectives; **theory** asks whether we specified the right one.

Rare-event example

Prevalence 0.1%; sensitivity = specificity = 99%; 100,000 cases

	Predict +	Predict –
Actual +	99 True Positive	1 False Negative
Actual –	999 False Positive	98,901 True Negative

- The classifier achieves 99% accuracy on positive and negative examples
- Only $99/(99 + 999) \approx 9\%$ of positive predictions are correct

AI Makes Search Cheap; Validation Data Remains Finite

Suppose an AI generates M candidate models and then choose the best using n holdout examples

We can prove:

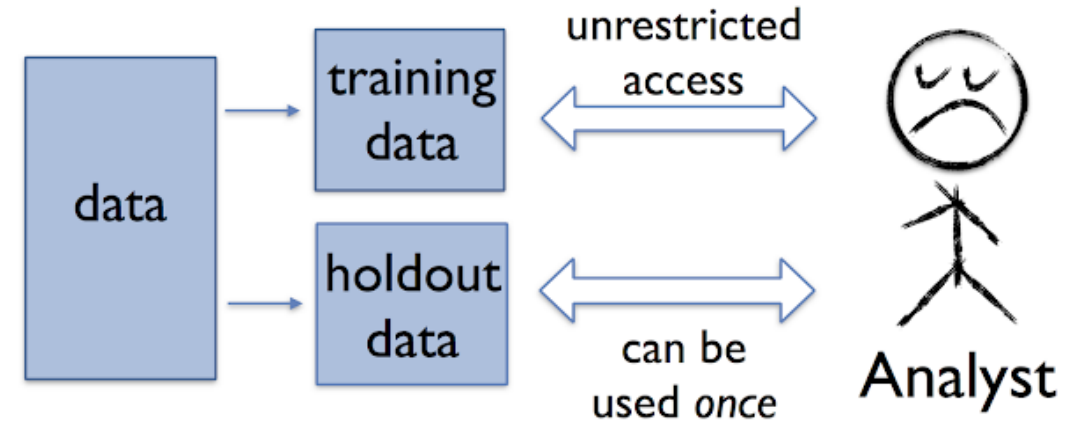
$$\max_{1 \leq i \leq M} |\mathcal{R}(f_i) - \hat{\mathcal{R}}_{\text{val}}(f_i)| \leq O\left(\sqrt{\log(M)/n}\right)$$

Population risk

Empirical risk

More candidates **increase** the statistical cost of selection

Standard holdout method



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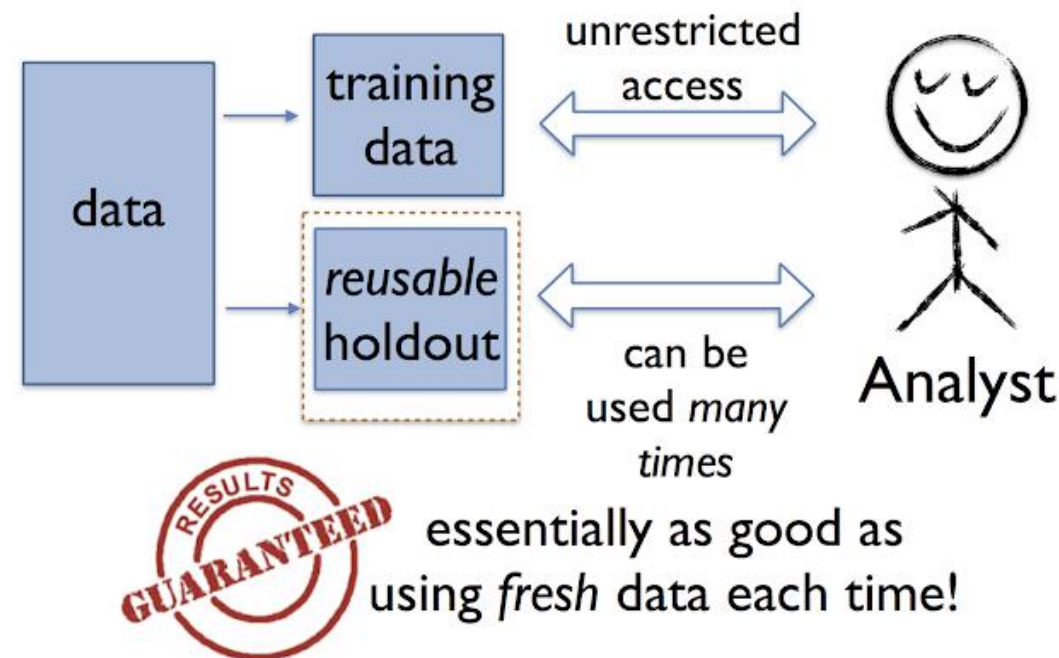
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Empirical risk

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Statistical theory can diagnose failure and supply a mechanism that changes the pipeline

Reusable holdout method



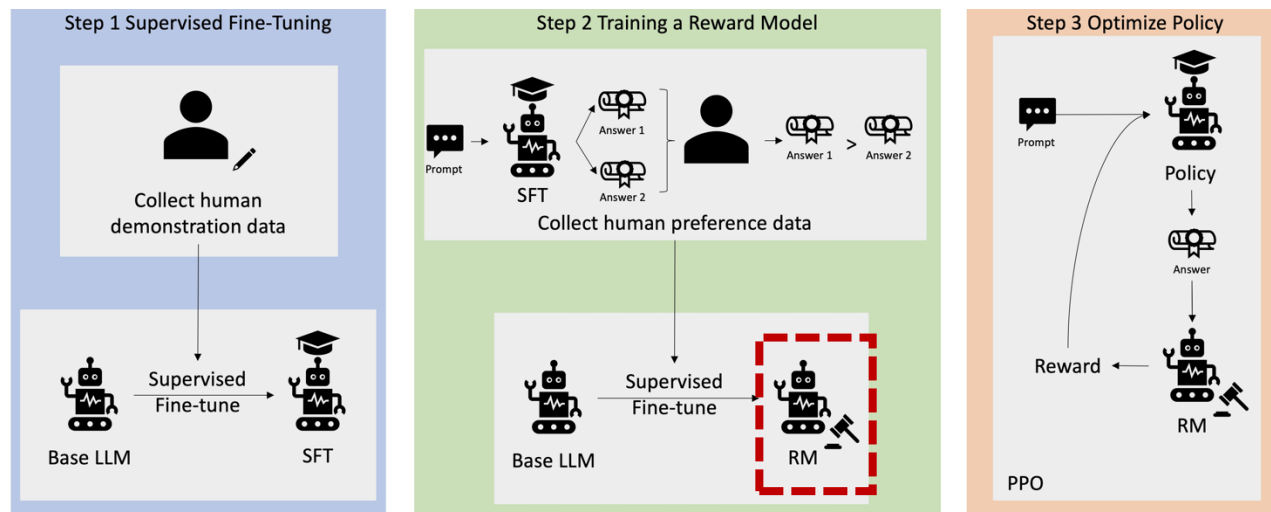
The reusable holdout: Preserving validity in adaptive data analysis

CYNTHIA DWORK, VITALY FELDMAN, MORITZ HARDT, TONIANN PITASSI, OMER REINGOLD, AND AARON ROTH [Authors Info & Affiliations](#)

SCIENCE • 7 Aug 2015 • Vol 349, Issue 6248 • pp. 636-638 • DOI: 10.1126/science.aaa9375

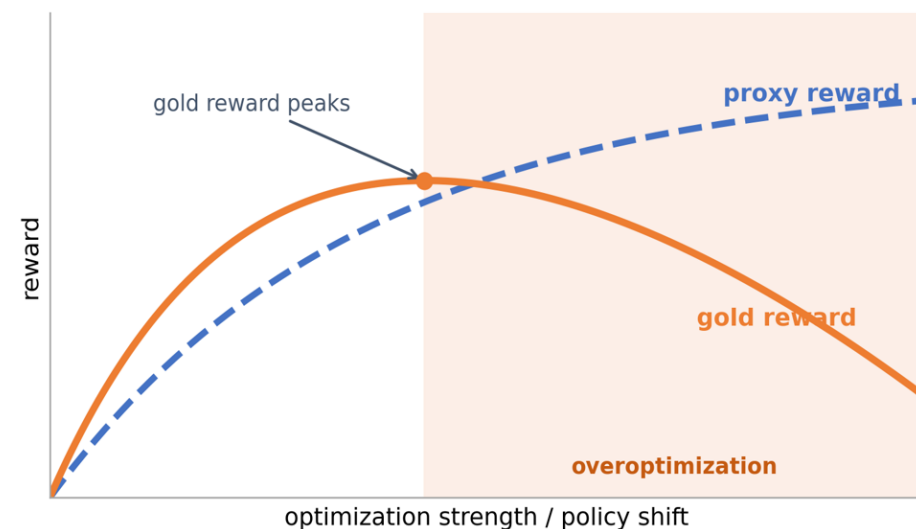
Optimization Can Exploit Errors in a Learned Objective

RLHF workflow



A reward model (RM) is only a proxy for true human values

Reward model overoptimization

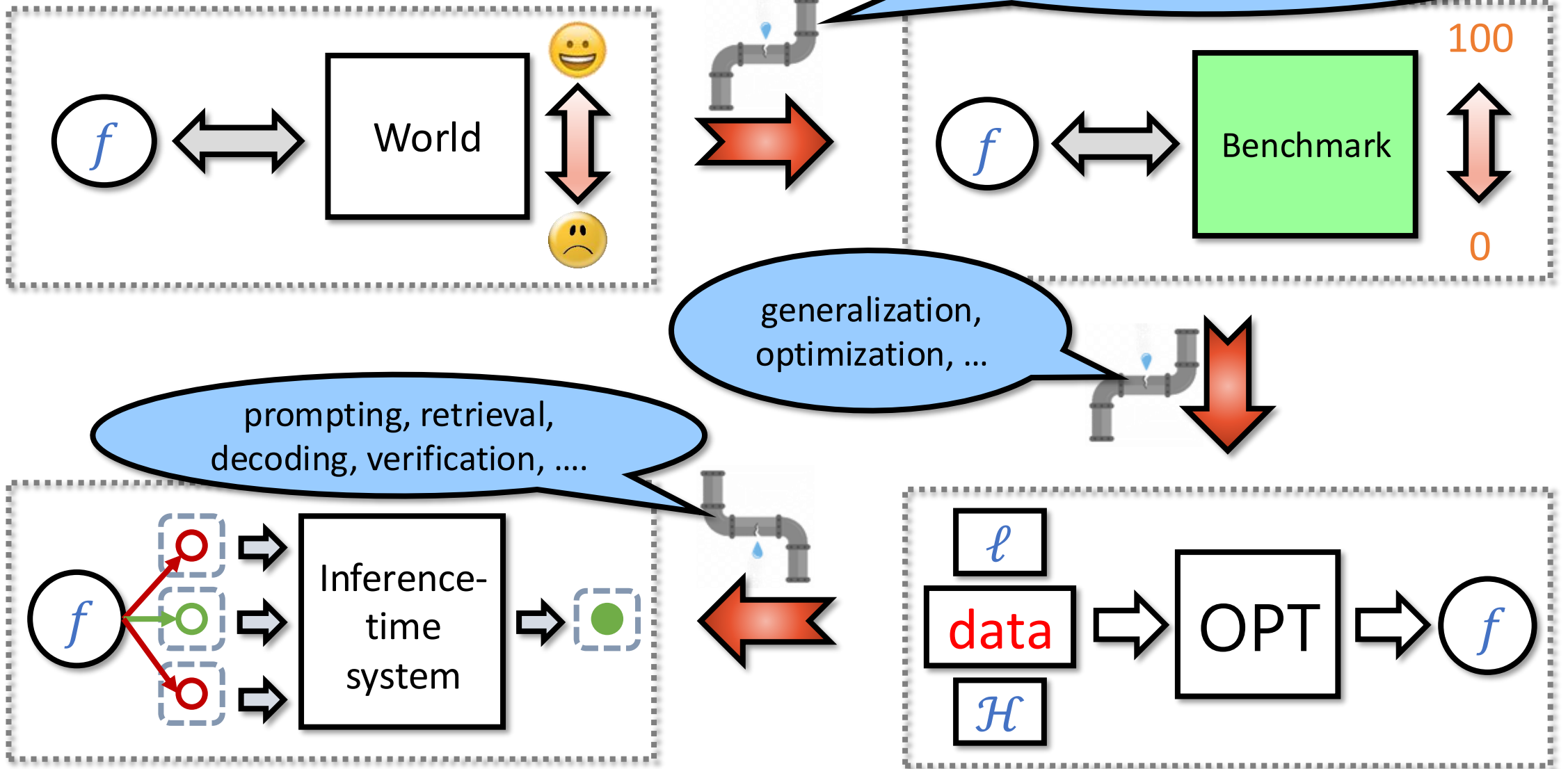


Scaling Laws for Reward Model Overoptimization

Leo Gao, John Schulman, Jacob Hilton *Proceedings of the 40th International Conference on Machine Learning*, PMLR 202:10835-10866, 2023.

A stronger optimizer can amplify objective misspecification.

Leaky Pipelines for ML



Adapted from Boaz Barak, "A blitz through classical statistical learning theory" (2021).

What This Course Is About

GENERALIZATION • INFERENCE • GENERATION • RELIABILITY

Statistical learning theory

PAC learning, ERM, Linear regression, ...

Unsupervised and latent-variable learning

Clustering, method of moments, identifiability, ...

Probabilistic inference and statistical physics

Energy-based models, variational methods, MCMC, mean-field, ...

Diffusion-model theory

Langevin dynamics, score estimation, SDE discretization, ...

Modern neural-network theory

NTK, edge of stability, in-context learning, ...

Trustworthy ML

Watermarking, alignment, differential privacy, ...

Concluding Remarks

- This course will be challenging; We are here to work together
- We will collaboratively learn from each other
- **Don't hesitate to stop me at any point to ask questions**